



Note

Assessment of clay consistency through conventional methods and indirect extrusion tests

R.D. Verástegui-Flores^{a,*}, G. Di Emidio^b^a iMMC, Université catholique de Louvain, Belgium^b Laboratory of Geotechnics, Ghent University, Belgium

ARTICLE INFO

Article history:

Received 29 August 2014

Accepted 20 September 2014

Available online 7 October 2014

Keywords:

Clay

Consistency

Liquid limit

Plastic limit

Extrusion

ABSTRACT

The consistency of fine-grained soil is an essential parameter in soil mechanics. The most relevant consistency indices are the liquid limit and the plastic limit. The liquid limit is commonly determined through the Casagrande test or the fall cone test, while the plastic limit is determined through the hand rolling method. The greatest issue with some of these techniques is their low repeatability and operator dependency. In order to minimize those issues, an indirect-extrusion-based technique was evaluated as an alternative method to determine both consistency limits. The experimental work was carried out on mixtures of kaolin and bentonite to cover a wide range of plasticity. The results suggested that there is a specific extrusion pressure linked to each consistency limit and that the results are repeatable. The liquid limit obtained through the extrusion method closely matches the results of the fall cone test. Similarly, the plastic limit out of extrusion closely matches the results of the hand rolling method.

© 2014 Elsevier B.V. All rights reserved.

1. Introduction

In soil mechanics, consistency is a fundamental parameter for the classification of fine-grained soil. The most relevant consistency indices are the liquid limit (LL) and the plastic limit (PL). Conventionally, LL is determined through the Casagrande test or the fall cone test, while PL is determined through the hand rolling method. The greatest issue with some of these techniques is their low repeatability and operator dependency. In order to minimize those issues, an indirect-extrusion-based technique is evaluated as an alternative method to determine both, LL and PL.

Extrusion is the mechanical production process by which a block of a material (billet) is reduced in cross section by forcing it to flow through a die orifice (Shi-Bo and Jian-Xin, 2007; Muenstermann et al., 2009; Zhang et al., 2011). Extrusion can be direct or indirect as illustrated in Fig. 1. Under direct extrusion (Fig. 1a) the direction of sample flow and ram travel is the same. Initially, the extrusion pressure rises rapidly up to a peak value. Then as extrusion initiates, the pressure decreases as the wall friction component decreases. Finally, at the billet's end, the pressure increases once again as the billet becomes short and the material exhibits more resistance to exit through the die. Conversely, under indirect extrusion (Fig. 1b), the die moves against the sample. In this way, no frictional forces are mobilized.

The concept of extrusion was probably first applied for geotechnical purposes by Whyte (1982); however, his work was focused on establishing a relationship between undrained shear strength, extrusion pressure and water content. Kayabali and Tufenkci (2010) investigated the indirect extrusion to determine the consistency limits of different types of natural soil with plasticity index ($PI = LL - PL$) below 50, taking as reference the consistency limit tests described by ASTM D 4318 (2010). However, the reported correlation between consistency limits and extrusion pressure showed some spreading probably due to the impact of friction in their testing setup.

The aim of this paper is to investigate whether it is possible to establish a stronger correlation for the determination of consistency limits of clay through the indirect extrusion technique by minimizing all possible sources of spread. At the same time, the results are compared with test methods described in ASTM D 4318 (2010) and BS 1377-2 (1990).

2. Materials

Tests were carried out on artificial samples consisting of mixtures of different percentages of kaolin (K) and bentonite (B) to cover a wide range of plasticity (i.e., 100 K, 80 K + 20B, 60 K + 40B, 40 K + 60B, 20 K + 80B, 100B). Some properties of the clay used in this research are given in Table 1.

The clay specimens were mixed with different amounts of deionized water in a dough mixer for about 15 min. Then the wet samples were stored in airtight containers at about 20 °C for a period of over 1 day. ASTM D4318-10 suggests a minimum curing period of 16 h before testing.

* Corresponding author at: iMMC, Université catholique de Louvain, Place du Levant 1, B-1348 Louvain-la-Neuve, Belgium. Tel.: +32 10 47 3020.

E-mail address: ramiro.verastegui@uclouvain.be (R.D. Verástegui-Flores).

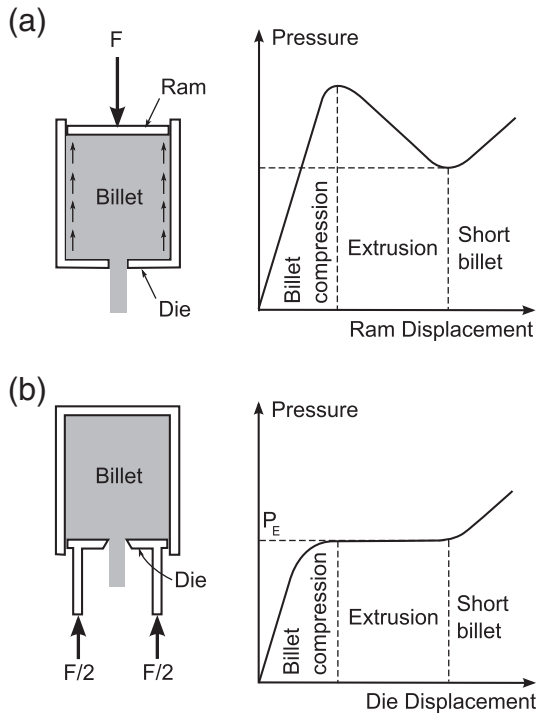


Fig. 1. Extrusion processes: (a) direct and (b) indirect.

The testing program on these samples included the execution of conventional tests to evaluate LL and PL and, more importantly, the development and evaluation of an indirect extrusion method. The outcome of the conventional tests serves as a reference for the calibration of the extrusion testing method.

3. Conventional testing methods

3.1. Evaluation of liquid limit

The liquid limit may be determined through the Casagrande cup test (ASTM D 4318, 2010) or the fall cone test (BS, 1377-2, 1990). The repeatability of the Casagrande cup method is known to be affected by operator judgment (Whyte, 1982; Özer, 2009). On the contrary, repeatability is not an issue for the fall cone test; however, the outcome of this test has to be carefully assessed as it will produce results for all soil types including non plastic soils, giving a wrong representation of plasticity (Prakash and Sridharan, 2006). The Casagrande cup and fall cone test do not produce the same results (Wasti, 1987; Schmitz et al., 2004; Prakash and Sridharan, 2006; Lee and Freeman, 2009; Özer, 2009; Zentar et al., 2009). In general, good agreement is observed for soils within an approximate range of $0\% < LL < 70\%$. Outside this range, the fall cone test produces consistently lower LL values.

3.2. Evaluation of plastic limit

PL is evaluated out of the standardized hand rolling method (BS, 1377-2, 1990; ASTM D 4318, 2010). However, several fall cone-based

methods are available in the literature (Wood and Wroth, 1978; Wasti, 1987; Feng, 2004; Lee and Freeman, 2009). In this research, the method proposed by Feng (2004) was followed as well. Here, four fall cone measurements are carried out to obtain cone penetrations ranging from 10 mm to 3 mm. Then PL is determined, through linear regression of the obtained data in a bi-logarithmic chart, as the water content corresponding to a cone penetration of 2 mm.

4. Indirect extrusion testing setup

The equipment used for the indirect extrusion tests consists of a load frame provided with a platen that moves at adjustable constant rates, an indirect extrusion device (Fig. 2), a load cell with a capacity of 2 kN and a displacement transducer. The indirect extrusion device consists of a steel cylindrical container and a steel extrusion die. The die orifice with diameter d has a conical shape to minimize contact friction between the extruded sample and the die. The tube behind the die shows a length of about 110 mm to allow enough space for a soil thread to develop without interruptions. During a test, the die is set to travel against the clay sample at a constant rate v while the die displacement and the extrusion load are recorded.

The extrusion pressure (P_E) of a clay sample at a specific water content is evaluated out of the steady-state portion of the measured extrusion load vs. die displacement curve (e.g. Fig. 1b) through

$$P_E = \frac{F_E}{A} \quad (1)$$

where F_E is the steady-state extrusion load and A is the cross section of the sample container.

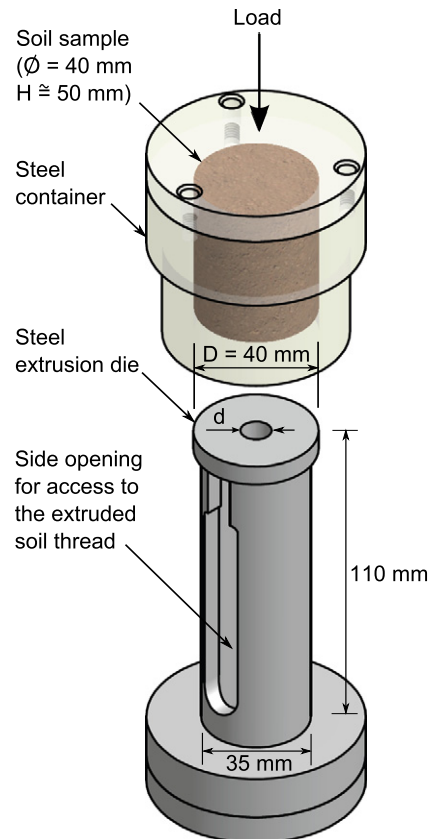


Fig. 2. Indirect extrusion device used in this research.

Table 1
Physical and chemical properties of the clay materials.

Index		Bentonite	Kaolin
Specific gravity	ASTM D 854	2.52	2.64
Liquid limit, %	ASTM D 4318	374.7	53.2
Plastic limit, %	ASTM D 4318	62.9	31.0
Swell index, ml/2 g	ASTM D 5890	18	3.5
CEC, mEq/100 g		73.77	1.38

P_E depends not only on the water content of the clay but also on the rate of die displacement (v) and on the extrusion factor (r). The extrusion factor is defined here as

$$r = \frac{D}{d} \quad (2)$$

where D is the diameter of the sample container and d is the diameter of the die orifice.

In order to evaluate the impact of r and v on P_E , a number of tests were carried out on a kaolin sample at its LL. Three die types, with extrusion factors $r = 2, 4$ and 8 , were tested. For each die, tests were carried out at $v = 2, 4, 6$ and 8 mm/min. The results (Fig. 3) show that, overall, P_E increases with increasing v . Tests with $r = 8$ show that P_E significantly increases with increasing v up to $v = 6$ mm/min. With $r = 4$, P_E is shown to significantly increase with increasing v up to $v = 4$ mm/min. Beyond this rate, the relative increase of P_E is less pronounced. With $r = 2$, the increase of P_E with v seems to be continuous. Moreover, for any v value, P_E increases with increasing r , as more energy is needed for the extrusion.

For the sake of simplicity, it is desired that an indirect extrusion test is performed making use of a single extrusion factor and a single die displacement rate that produce representative and reproducible data. Hence, a choice was made taking into account the previous remarks. The die with $r = 4$ and a rate $v = 4$ mm/min were selected as the most suitable for all further indirect extrusion tests in this research.

5. Results and discussion

5.1. Conventional methods

The consistency limits out of conventional tests are summarized in Table 2. As expected, LL and PL increase with increasing bentonite content. In agreement with the literature, LL out of the Casagrande cup and fall cone tests show similar values only for the lower range of plasticity. PL was determined out of the hand method and through the fall cone method (Feng, 2004) and the results show good agreement.

5.2. Indirect extrusion

The testing program with the indirect extrusion technique included first the determination of the steady-state P_E values corresponding to LL

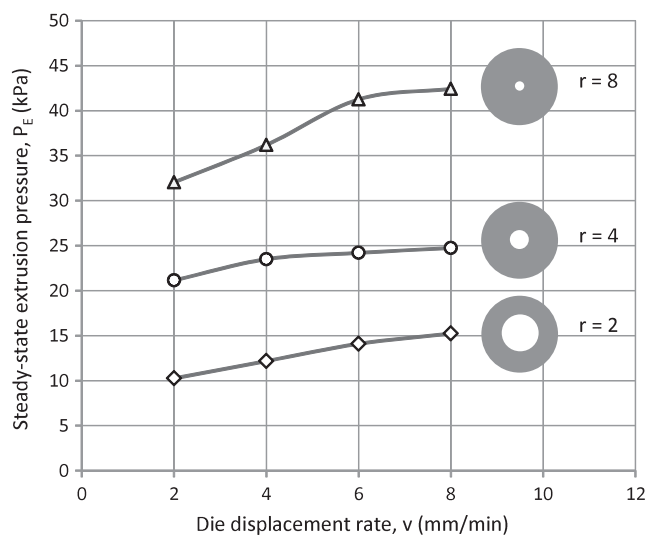


Fig. 3. Impact of the die displacement rate (v) and extrusion factor (r) on the steady-state extrusion pressure (P_E).

Table 2
Consistency limits out of conventional methods.

Sample	LL Casagrande	LL Fall cone	PL Hand	PL Fall cone
100 K	53.2	55.6	31.0	29.3
80 K + 20 B	86.5	76.8	34.8	29.6
60 K + 40 B	142.4	118.1	37.2	33.9
40 K + 60 B	208.1	168.8	48.6	45.9
20 K + 80 B	282.4	216.9	51.9	54.7
100 B	374.7	279.6	62.9	64.1

and PL on reference samples. Next, it follows the verification of such P_E values as indicators of consistency in other soil samples.

For the determination of a characteristic P_E value at LL, a sample of kaolin was taken as reference because its LL out of Casagrande and fall cone tests was rather similar ($LL_{Cas} = 53.2\%$ and $LL_{FC} = 55.6\%$). Then a kaolin sample was prepared at a water content equal to the average LL. Several indirect extrusion tests were carried out on this sample. Fig. 4a shows the recorded P_E vs. die displacement. The evaluation of the

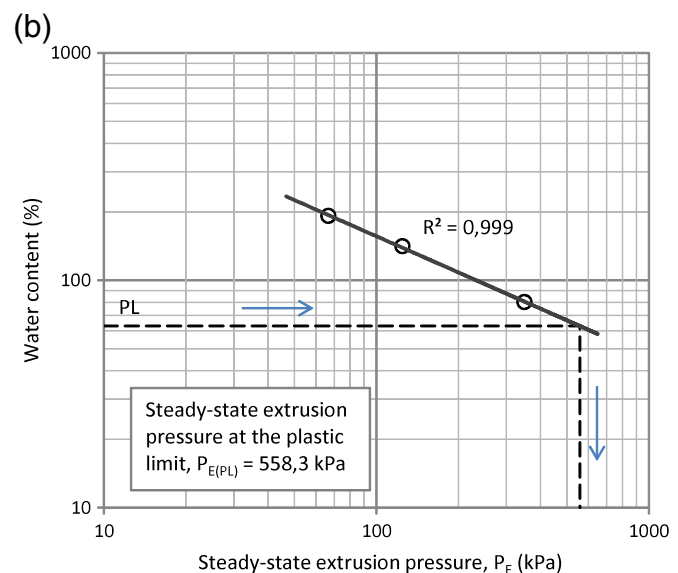
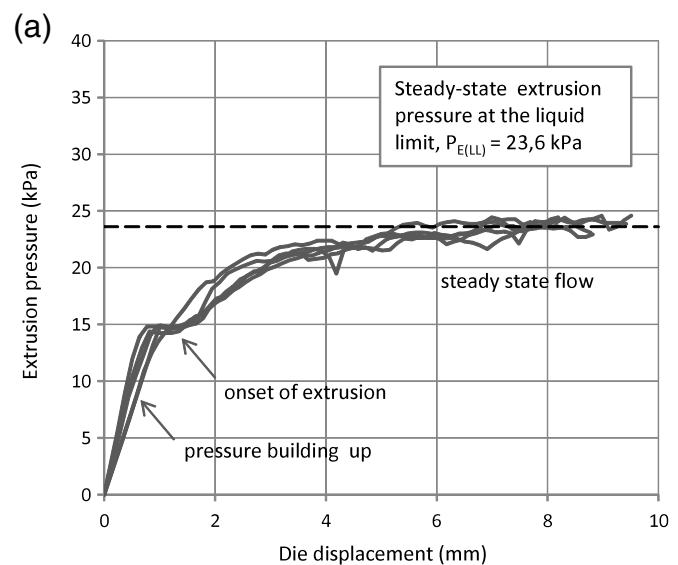


Fig. 4. Evaluation of the steady-state extrusion pressure at (a) the liquid limit state (b) the plastic limit state.

steady-state P_E value at the liquid limit state was straightforward, and it shows little spread with an average value of $P_{E(LL)} = 23.6$ kPa.

For the determination of a characteristic P_E value at PL, kaolin was initially taken as reference sample as well. Unlike samples at LL, samples at PL are hardly workable. Therefore, it was opted for performing tests on samples at water contents slightly higher than the plastic limit and evaluating P_E at PL out of regression and extrapolation as done in the fall cone method (Feng, 2004). However, indirect extrusion tests on kaolin clay showed unexpected features. The mobilized P_E did not reach a steady-state value; instead, it continuously increased. Moreover, the extruded clay thread showed an increased water content. Such phenomena suggested that the kaolin sample in the container was not only undergoing extrusion but also compression. Then pore water was being forced out of the container through the die opening. As a result, the density and the strength of the material in the container gradually increased leading to a continuous increase of P_E .

These results initially suggested that the extrusion technique was not applicable to low plasticity clays for the determination of PL. However, indirect extrusion tests on a sample of natural Boom clay (low plasticity clay containing illite) with $LL_{Cas} = 51.0\%$ and $PL_{hand} = 32.4\%$ (very similar to kaolin) showed no signs of dewatering and most importantly the extrusion pressure vs. die displacement curve showed a well-defined steady-state trend. In addition to that, Kayabali and Tufenkci (2010) tested about 30 natural soil samples (with $83\% > LL > 26\%$ and $47\% > PL > 18\%$) through an extrusion-based technique without reporting dewatering issues. Then the problem encountered here with tests on kaolin at water contents near PL was attributed to the specific mineralogical characteristics of the industrially purified kaolin sample used in this research. In general (among kaolinite, illite and smectite), kaolinite shows high hydraulic conductivity and low specific surface (Mesri and Olson, 1971), so it is more prone to release water when subjected to stress.

Then it was decided to use bentonite as reference material for the determination of the characteristic P_E value at PL. The relationship between water content (w) of the sample and steady-state P_E was found to be non-linear in a semi-logarithmic chart. However, when plotting the same points in a bi-logarithmic chart, a very well-defined linear trend could be observed (Fig. 4b). Finally, the extrusion pressure corresponding to PL of the reference sample ($PL_{hand} = 62.9\%$) was determined as $P_{E(PL)} = 558.3$ kPa.

Next, to verify the validity of $P_{E(LL)}$ and $P_{E(PL)}$ as indicators of consistency, indirect extrusion tests were carried out on the rest of the samples consisting of mixtures of kaolin and bentonite. The followed procedures are described below.

For the determination of LL, 3 to 4 indirect extrusion tests are carried out on samples at different water contents (w) to produce P_E values within a range of approximately 10 kPa to 50 kPa. The relationship between w of the extruded soil thread and P_E should show a rather linear trend in a semi-logarithmic chart. Then LL is defined through linear interpolation as the water content corresponding to $P_{E(LL)} = 23.6$ kPa.

For the determination of PL, 3 to 4 indirect extrusion tests are carried out on samples at different water contents to produce P_E values within a range of approximately 100 kPa to 400 kPa. Then the relationship w vs. P_E is plotted in a bi-logarithmic chart, and finally, PL is defined through linear extrapolation as the water content corresponding to $P_{E(PL)} = 558.3$ kPa.

In Fig. 5, the estimated consistency limits of all samples are summarized and compared to the outcome of conventional tests. LL values out of indirect extrusion tests are lower than values out of Casagrande cup tests. However, a very good match was observed with fall cone test results. A reason for this could lie in the different nature of these tests. The fall cone test and the indirect extrusion test subject the clay sample to monotonic loading, whereas a sample in the Casagrande cup is subjected to an instantaneous (dynamic) load when the cup is dropped. Because the strength of soil (especially high plasticity clays) is sensitive to loading rate effects (Diaz-Rodríguez et al., 2009), the sample will respond to the instantaneous load with a strength magnitude higher

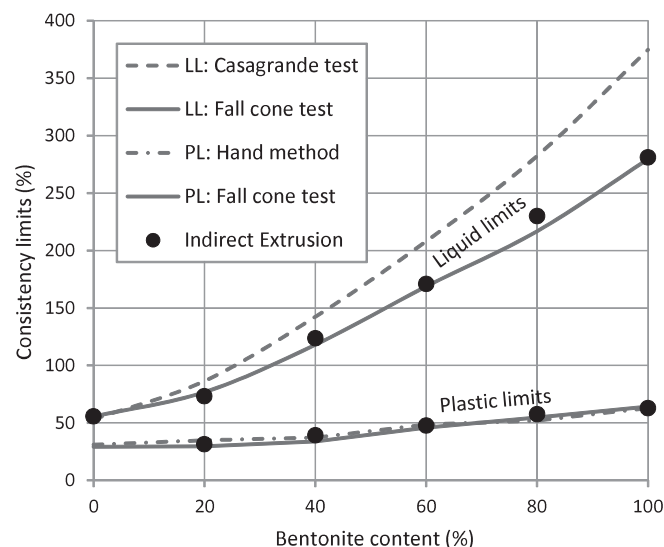


Fig. 5. Consistency limits from indirect extrusion vs. conventional tests.

than that under monotonic conditions. Therefore, the Casagrande test may allow for higher water contents on the clay sample.

Finally, PL estimated out of indirect extrusion shows good agreement with the results obtained out of the conventional methods.

6. Conclusions

An indirect-extrusion-based method was evaluated as an alternative technique to determine LL and PL of clay. The chosen extrusion factor of the die and the die displacement rate are $r = 4$ and $v = 4$ mm/min, respectively. During the indirect extrusion process, the mobilized extrusion pressure is continuously monitored vs. die displacement. Out of these measurements, a steady-state extrusion pressure (P_E) is evaluated. This test is repeated several times at various water contents to produce P_E values approximately between 10 kPa to 50 kPa for the determination of LL and between 100 kPa to 400 kPa for the determination of PL. The liquid limit is defined through linear interpolation (in a semi-logarithmic chart) as the water content corresponding to an extrusion pressure of $P_{E(LL)} = 23.6$ kPa. The plastic limit is defined through linear extrapolation (in a bi-logarithmic chart) as the water content corresponding to an extrusion pressure of $P_{E(PL)} = 558.3$ kPa.

The liquid limits estimated out of indirect extrusion tests are lower than those out of Casagrande tests; however, a very good match was observed with liquid limits out of fall cone testing. The plastic limit estimated out of indirect extrusion shows good agreement with the results obtained out of the conventional hand method and the fall cone-based method implemented here. An important advantage of the indirect extrusion technique is that it is able to produce both consistency limits while operator judgment plays a less important role.

References

- ASTM D 4318, 2010. Standard test methods for liquid limit, plastic limit, and plasticity index of soils. ASTM International.
- ASTM D 5890, 2011. Standard test method for swell index of clay mineral component of geosynthetic clay liners. ASTM International.
- ASTM D 854, 2010. Standard test methods for specific gravity of soil solids by water pycnometer. ASTM International.
- BS 1377-2, 1990. Methods of test for soils for civil engineering purposes. Classification tests. British Standards Institution.
- Diaz-Rodríguez, J.A., Martínez-Vasquez, J.J., Santamarina, J.C., 2009. Strain rate effects in Mexico City soil. *J. Geotech. Geoenviron.* 135 (2), 300–305.
- Feng, T.W., 2004. Using small ring and a fall cone to determine the plastic limit. *J. Geotech. Geoenviron.* 130 (6), 630–635.
- Kayabali, K., Tufenkci, O.O., 2010. Determination of plastic and liquid limits using the reverse extrusion technique. *Geotech. Test. J.* 33 (1), 14–22.

- Lee, L.T., Freeman, R.B., 2009. Dual-weight fall cone method for simultaneous liquid and plastic limit determination. *J. Geotech. Geoenviron.* 135 (1), 158–161.
- Mesri, G., Olson, R.E., 1971. Mechanisms controlling the permeability of clays. *Clay Clay Miner.* 19, 151–158.
- Muenstermann, S., Uibel, K., Tonnesen, T., Telle, R., 2009. Semi-solid extrusion of steel grade X210CrW12 under isothermal conditions using ceramic dies. *J. Mater. Process. Technol.* 209 (7), 3640–3649.
- Özer, M., 2009. Comparison of liquid limit values determined using the hard and soft base Casagrande apparatus and the cone penetrometer. *Bull. Eng. Geol. Environ.* 68 (3), 289–296.
- Prakash, K., Sridharan, A., 2006. Critical appraisal of the cone penetration method of determining soil plasticity. *Can. Geotech. J.* 43, 884–888.
- Schmitz, R.M., Schroeder, C., Charlier, R., 2004. Chemo-mechanical interactions in clay: a correlation between clay mineralogy and Atterberg limits. *Appl. Clay Sci.* 26, 351–358.
- Shi-Bo, L., Jian-Xin, X., 2007. Fabrication of thin-walled 316 L stainless steel seamless pipes by extrusion technology. *J. Mater. Process. Technol.* 183 (1), 57–61.
- Wasti, Y., 1987. Liquid and plastic limits as determined from the fall cone and the Casagrande methods. *Geotech. Test. J.* 10 (1), 26–30.
- Whyte, I.L., 1982. Soil plasticity and strength—a new approach using extrusion. *Ground Eng.* 15 (1), 16–24.
- Wood, D.M., Wroth, C.P., 1978. The use of the cone penetrometer to determine the plastic limit of soils. *Ground Eng.* 11 (3), 37.
- Zentar, R., Abriak, N.E., Dubois, V., 2009. Effects of salts and organic matter on Atterberg limits of dredged marine sediments. *Appl. Clay Sci.* 42 (3–4), 391–397.
- Zhang, C., Zhao, G., Chen, Z., Chen, H., Kou, F., 2011. Effect of extrusion stem speed on extrusion process for a hollow aluminum profile. *Mater. Sci. Eng.* <http://dx.doi.org/10.1016/j.mseb.2011.09.041>.